(IoT 스마트카 시스템) 주간진행보고서

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1 주간 진행 내용 (12/01~12/07)

구현 내용, H/W 사진, S/W 스케치 등
- 블루투스센서 이용하여 스마트카 자동 소환 및 자동 주차
- 어두워지면 자동으로 전조등 켜기
- 주행 중 장애물 인지 시 경적 울리며 멈추기
- 라인트레이서 센서를 이용하여 도로 주행

```
#include <Servo.h>
#include <SoftwareSerial.h>
SoftwareSerial BT(5, 6);
Servo servoLeft;
Servo servoRight;
#define LEFT_S    2
#define RIGHT_S   4
#define TRIG_F    8
#define ECHO_F    9
#define TRIG_B    10
#define ECHO_B    11
#define LED_L A0
#define LED_R A1
#define LDR    A2
#define BUZZER A3
const int SPEED_FWD = 60;
const int SPEED_TURN_FAST = 80;
const int SPEED_TURN_ADJUST = 50;
const int OBSTACLE_DIST = 8;
const int PARKING_DIST = 7;
int traceMode = 0;
void stopMove() {
  servoLeft.writeMicroseconds(1500);
  servoRight.writeMicroseconds(1500);
}long getDistance() {
  digitalWrite(TRIG_F, LOW); delayMicroseconds(2);
  digitalWrite(TRIG_F, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG_F, LOW);
  long duration = pulseIn(ECHO_F, HIGH);
```

```
  return duration * 0.034 / 2;
}long getBackDistance() {
  digitalWrite(TRIG_B, LOW); delayMicroseconds(2);
  digitalWrite(TRIG_B, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG_B, LOW);
  long duration = pulseIn(ECHO_B, HIGH);
  return duration * 0.034 / 2;
}void maneuver(int left, int right, int ms) {
  servoLeft.writeMicroseconds(1500 + left);
  servoRight.writeMicroseconds(1500 - right);
  if (ms > 0) delay(ms);
}void runSmartTracing() {
  int L = digitalRead(LEFT_S);
  int R = digitalRead(RIGHT_S);
  long distance = getDistance();
  if (distance > 0 && distance <= OBSTACLE_DIST) {
    stopMove();
    tone(BUZZER, 2000);
    delay(100);
    noTone(BUZZER);
    return;
  } else {
    noTone(BUZZER);
  }
  if (traceMode == 0) {
    if (L == 0) traceMode = 1;
    else if (R == 0) traceMode = 2;
    else maneuver(SPEED_FWD, SPEED_FWD, 20);
  }
  else if (traceMode == 1) {
    if (R == 0) {
      traceMode = 2;
      maneuver(0, 0, 100);
      maneuver(SPEED_TURN_FAST, -80, 400);
    }
    else if (L == 0) maneuver(SPEED_FWD, SPEED_FWD, 20);
    else maneuver(0, SPEED_TURN_ADJUST, 20);
  }
  else if (traceMode == 2) {
    if (L == 0) {
      traceMode = 1;
      maneuver(0, 0, 100);
```

```
      maneuver(-80, SPEED_TURN_FAST, 400);
    }
    else if (R == 0) maneuver(SPEED_FWD, SPEED_FWD, 20);
    else maneuver(SPEED_TURN_ADJUST, 0, 20);
  }
}void autoHeadlight() {
  int lightVal = analogRead(LDR);
  if (lightVal < 300) {
    analogWrite(LED_L, 180);
    analogWrite(LED_R, 180);
  } else {
    analogWrite(LED_L, 0);
    analogWrite(LED_R, 0);
  }
}void setup() {
  Serial.begin(9600);
  BT.begin(9600);
  pinMode(LEFT_S, INPUT);
  pinMode(RIGHT_S, INPUT);

  pinMode(TRIG_F, OUTPUT);
  pinMode(ECHO_F, INPUT);

  pinMode(TRIG_B, OUTPUT);
  pinMode(ECHO_B, INPUT);
  pinMode(LED_L, OUTPUT);
  pinMode(LED_R, OUTPUT);
  pinMode(BUZZER, OUTPUT);
  servoLeft.attach(13);
  servoRight.attach(12);
  stopMove();
  tone(BUZZER, 2000, 100); delay(150);
  tone(BUZZER, 3000, 100);
  Serial.println("ABot Ready: Send S, D, P, X");
}void loop() {
  autoHeadlight();
  if (BT.available()) {
    char cmd = BT.read();
    if(cmd == '\n' || cmd == '\r') return;
    if (cmd == 'X') {
      stopMove();
      Serial.println("System Stopped");
```

```
  tone(BUZZER, 1000, 200);
}
else if (cmd == 'S') {
  Serial.println("Summon Mode START!");
  traceMode = 0;
  tone(BUZZER, 3000, 100);
  while (true) {
    autoHeadlight();

    long dist = getDistance();
    if (dist > 0 && dist < 10) {
      stopMove();
      Serial.println("Summon Complete!");
      for(int i=0; i<3; i++) { tone(BUZZER, 2000, 100); delay(150); }
      break;
    }
    runSmartTracing();
    if (BT.available()) {
      if (BT.read() == 'X') {
        stopMove();
        Serial.println("Summon Cancelled");
        break;
      }
    }
  }
}
else if (cmd == 'D') {
  Serial.println("Drive Mode START!");
  traceMode = 0;
  tone(BUZZER, 3000, 100);
  while (true) {
    autoHeadlight();
    runSmartTracing();
    if (BT.available()) {
      if (BT.read() == 'X') {
        stopMove();
        Serial.println("Drive Stopped");
        break;
      }
    }
  }
}
```

```
else if (cmd == 'P') {
  Serial.println("Auto Parking (Back-Left Curve) START!");
  tone(BUZZER, 3000, 100);

  Serial.println("Step 1: Curve Back Left...");

  long startTime = millis();
  while (true) {
    long b_dist = getBackDistance();
    Serial.print("Back Dist: "); Serial.println(b_dist);
    if (b_dist > 0 && b_dist < 2) {
      stopMove();
      delay(500);
      break;
    }
    if (millis() - startTime > 2000) {
      stopMove();
      Serial.println("Timeover! Check Position.");
      tone(BUZZER, 500, 1000);
      break;
    }
    maneuver(-30, -80, 0);
    delay(50);

    if (BT.available() && BT.read() == 'X') { stopMove(); return; }
  }
  Serial.println("Step 2: Wari-gari with Steering (5 times)");

  for (int i = 1; i <= 3; i++) {
    Serial.print("Adjustment Round: "); Serial.println(i);

    maneuver(40, 60, 400);
    stopMove();
    delay(200);
    while(true) {
      long b_dist = getBackDistance();
      if (b_dist > 0 && b_dist <= PARKING_DIST) {
        stopMove();
        break;
      }
      maneuver(-35, -35, 0);
      delay(50);
```

```
          if (BT.available() && BT.read() == 'X') { stopMove(); return; }
        }
        delay(200);
      }
      Serial.println("Parking Complete!");
      for(int i=0; i<3; i++) {
        analogWrite(LED_L, 255); analogWrite(LED_R, 255);
        tone(BUZZER, 2000); delay(200);
        analogWrite(LED_L, 0); analogWrite(LED_R, 0);
        noTone(BUZZER); delay(200);
      }
    }
  }
}
```

2 진행 중 어려운 점, 문의 사항

없음.

3 다음주 계획 (11/17~11/23)

- IoT 스마트카 구현 동영상 촬영
- 최종 보고서 작성